

Justice, served digital

One interesting solution provided by the Big Data start-up SenziT, in association with IBM, is the use of live digital-recording methods, artificial intelligence and complex analytics software to provide content capturing, management, tracking and analyses solutions for department of justice and the judiciary.

Their products let the user capture live proceedings, right from the crime scene to police investigations to the court room, in the form of text and high-quality audio and video, on an enterprise-mobility platform. IBM's powerful Worklight software is used to create a highly-secure mobile framework. An artificially-intelligent computer can use its own algorithms to aggregate these different types of information feeds, correlate them and present the analysed data in a form that is easily understood by the viewer. With latest technologies like Big Data analytics and enterprise video analytics being used, this method could realise otherwise unnoticeable insights, patterns and correlations in the huge amount of evidence and data, and hence provide a powerful, predictive and visual-analysis platform.

"This method can also be used in smart classrooms, where the teacher can assess which student is less focussed or misbehaving in class," says Shibi Salim, director, SenziT, "and in medical applications, where the system suggests all possible medical conditions of a patient based on his or her symptoms and histories."

like anemometers (to measure wind speed and direction), pyranometers (to measure solar irradiance on planar surface), pressure sensors, temperature sensors and humidity sensors can help understand the potential of energy available for conversion. Sensors within the equipment's subsystems, such as generator, rotor system, gearbox, solar panels and inverters, give information on the electrical and mechanical performance of the system.

Also Engines interfaces with hundreds of different sensors, supervisory control and data acquisition (SCADA) systems and smart metres, and pulls information collected by them to the cloud, analyses this massive and complex data using IBM tools for better predictive maintenance and forecast. A web based platform is made available to the customer where they can gauge the health of the device, monitor its performance, and view forecasts and predictions of component failure. These factors further increase the efficiency, effectiveness and, hence, revenue of the system.

Life-sciences and medicine

Big Data analysis in IoT networks is taking the idiom 'prevention is better than cure' to a whole new level.

Advanced sensors, mobile apps and extensive databases are helping us reach the goal of personal health improvement.

Converting thoughts into actions. Brain-computer interface (BCI), also known as brain-machine interface (BMI) or mind-machine interface (MMI) or direct neural interface, is a method that conjoins the human brain and machines, and lets you control external devices like bionic modules using your thoughts. The system consists of several powerful and precise sensors (typically multi-electrode arrays) that collect brain signals, a decoder that can detect and interpret neural signals related to each activity, and an external module like a computer screen or robotic arm that translates these electronic signals into corresponding actions. This technology is a breakthrough in the field of medicine as it can help restore disabilities of patients, and institutions like BrainGate and Pitt School of Medicine (University of Pittsburgh) are putting in a lot of research in this field.

From existing to next-gen intelligent system

Converting from an old-fashioned system to a next-generation intelligent connected system is going to



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